

ROBOT PLATFORM

Rosie

A six-axis articulated robot platform powered by AMR's GradientOS, the open-source industrial robotics controller.

OPEN SOURCE ★ NO FEES ★ EXPANDABLE IO ★ HOLLOW ROUTING

OPEN CONTROLLER

Full industrial robotics controller, open source, with no hidden software fees.

EASY INTEGRATION

Open GradientOS stack reduces lock-in and keeps functionality accessible.

EXPANDABLE IO

Ready for cell devices, tooling, safety signals, and end-effector integration.

HOLLOW CABLE PATHS

62 mm shoulder and 33 mm wrist pass-throughs for cleaner cable routing.



Payload

15 kg

Nominal handling capacity for Rosie.

Horizontal Reach

1,509 mm

Current GradientOS kinematic value at the `tool0` origin.

Vertical Reach

1,638 mm

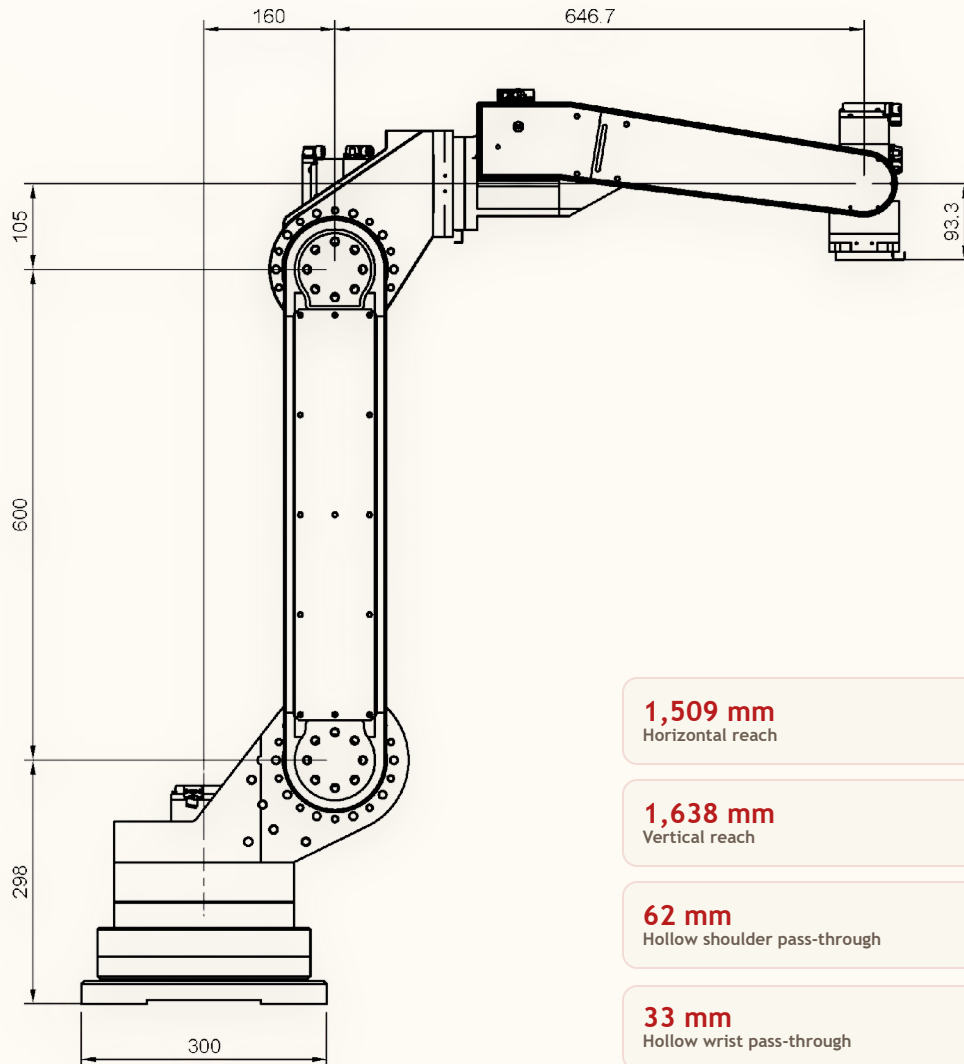
Above-base vertical reach; total envelope span is a separate sweep value.

Repeatability

0.08 mm

Published repeatability for Rosie.

Robot Dimensions



1,509 mm
Horizontal reach

1,638 mm
Vertical reach

62 mm
Hollow shoulder pass-through

33 mm
Hollow wrist pass-through

Axis Specifications

AXIS	ROLE	MOTION RANGE	RATED SPEED	PEAK SPEED
J1	Base	+/- 361.0 deg	180 deg/s	360 deg/s
J2	Shoulder	+/- 108.9 deg	180 deg/s	360 deg/s
J3	Elbow	-240.6 / +87.7 deg	180 deg/s	360 deg/s
J4	Wrist roll	+/- 361.0 deg	1,000 deg/s	2,000 deg/s
J5	Wrist pitch	+/- 361.0 deg	1,000 deg/s	2,000 deg/s
J6	Tool roll	+/- 573.0 deg	1,800 deg/s	3,600 deg/s

Open Controller

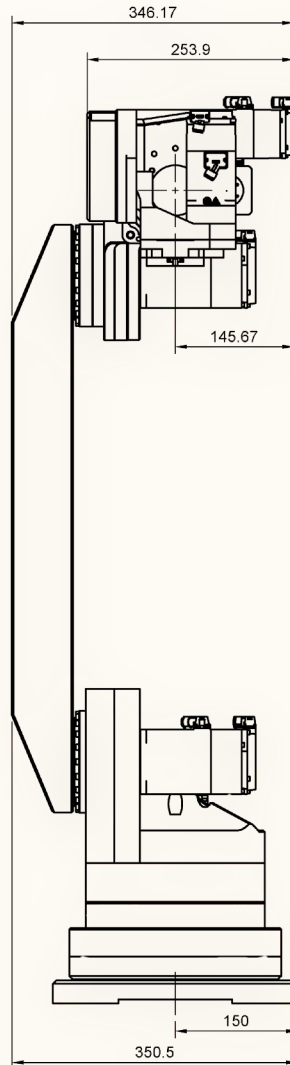
GradientOS is an open-source full industrial robotics controller, built to make integration simple without hidden software fees or artificially limited functionality.

Expandable IO supports cell devices, tooling, safety signals, and end-effector integration.

Controller Defaults

Robot ID	gradient-05	Gradient05Config
Physical actuators	6 EtherCAT axes	IDs 0-5
Profile velocity	0.1 m/s	default
Profile acceleration	0.05 m/s^2	default
Cartesian jog caps	0.2 m/s linear	180 deg/s angular
Encoder model	17-bit motor-side	absolute count model
Power	200-240 VAC 3PH	35A
Integration	Expandable IO	Open-source controller

Arm Envelope



346.17 mm
Wrist tool envelope

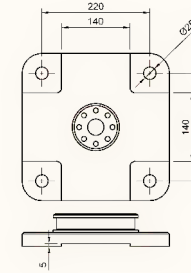
253.9 mm
Wrist housing span

145.67 mm
Tool-side offset

350.5 mm
Mount plate width

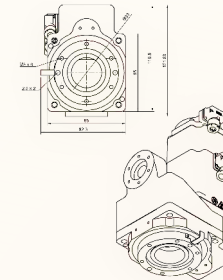
150 mm
Base column width

Base Mount



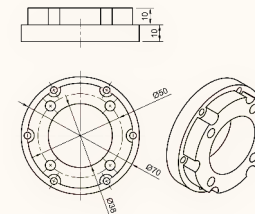
Mount plate	220 x 220 mm
Bolt circle	140 mm PCD
Anchor bore	Ø25.5 mm
Plate thickness	5 mm

Wrist Flange



Hollow bore	Ø33 mm
Bolt pattern	85 x 85 mm
Threaded mounts	M5 x 6
Flange width	92.3 mm

ISO50 Tool Adapter



Outer diameter	Ø70 mm
Bolt circle	Ø50 mm
Through bore	Ø38 mm
Stack height	10 mm